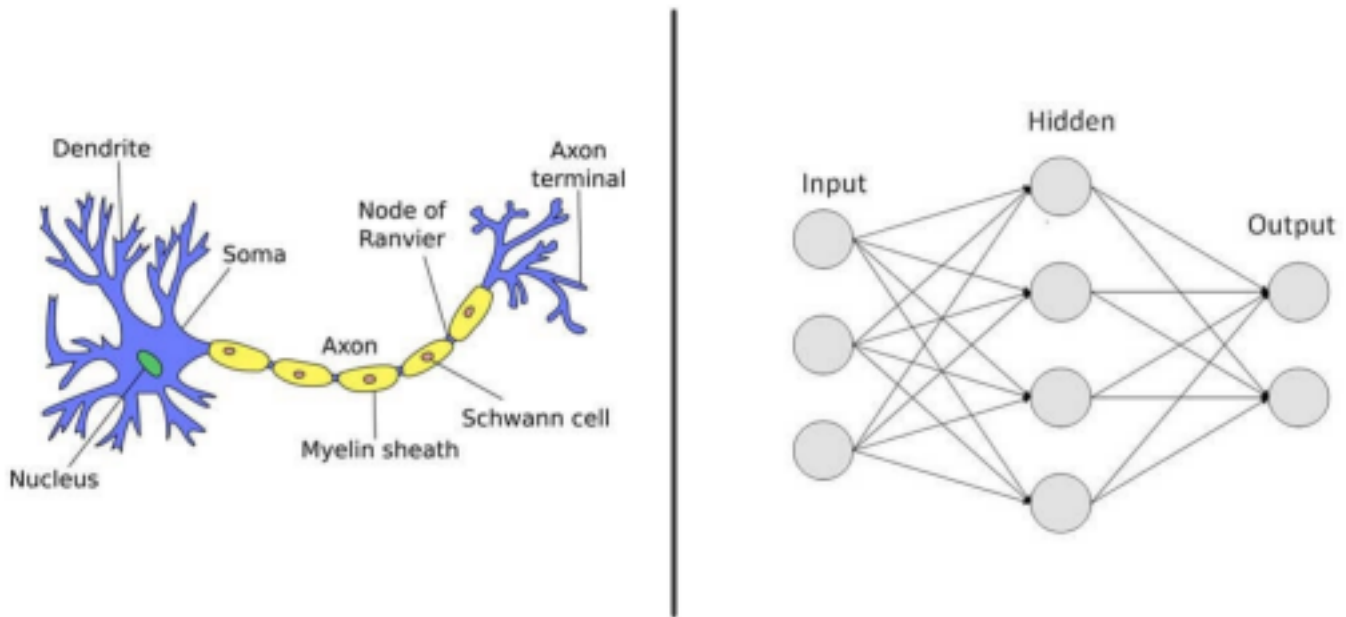


Biological Neural Network & Artificial Neural Network



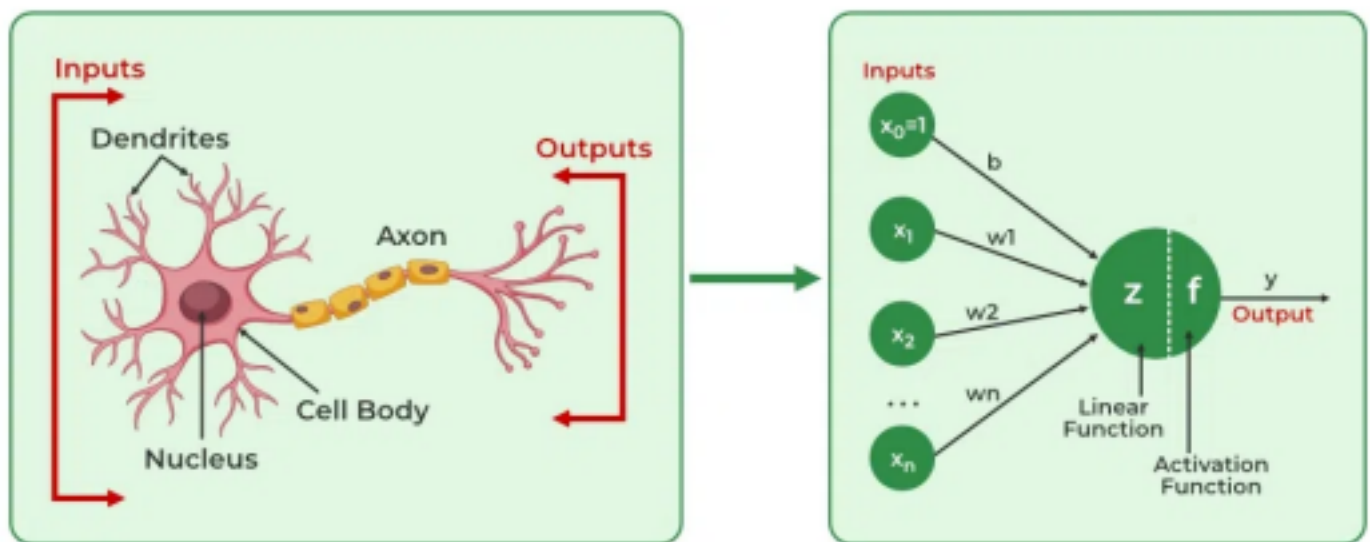
Biological Neural Network (BNN)

- Found in the **human brain and nervous system**.
- Consists of **billions of biological neurons** (brain cells).
- Each neuron is connected to thousands of others via **synapses**.
- Works on **electrochemical signals**.
- **Learning mechanism:** Adjusts the strength of synaptic connections over time (called **synaptic plasticity**).
- Extremely **parallel, adaptive, and energy-efficient**.

Artificial Neural Network (ANN)

- A **mathematical/computational model** inspired by the brain's BNN.
- Consists of **artificial neurons (perceptrons)** arranged in layers.
- Neurons process **numerical data**, not chemical signals.
- **Connections (weights):** Store learned importance of inputs.
- **Learning mechanism:** Adjusts weights using algorithms like **backpropagation** and **gradient descent**.

Runs on **CPUs/GPUs** with high power consumption.



Biological Neuron (Human Brain)

A **neuron** is the **basic unit of the brain** and nervous system. It processes and transmits information using **electrical and chemical signals**.

Structure

1. **Dendrites** – Branch-like structures that receive signals from other neurons.
2. **Cell Body (Soma)** – Processes the incoming signals.
3. **Axon** – A long extension that transmits signals away to other neurons.
4. **Synapse** – Junction where the axon of one neuron connects to the dendrite of another, using neurotransmitters.

Working

- **Input:** Signals arrive at dendrites.
- **Processing:** Soma integrates signals; if the total signal exceeds a threshold, the neuron “fires.”
- **Output:** Axon carries the electrical impulse to synapses.
- **Learning:** Repeated usage strengthens or weakens synaptic connections (**Synaptic Plasticity**).

Human brain has **~86 billion neurons**, each connecting to thousands of others → massive parallelism.

Artificial Neuron (Perceptron)

An **Artificial Neuron** is a **mathematical model** inspired by biological neurons. It takes inputs, processes them, and produces an output.

Structure

1. **Inputs ($x_1, x_2, \dots, x_n$):** Features/data points fed into the model.

2. **Weights (w1, w2, ..., wn):** Each input has an importance factor.
3. **Bias (b):** A constant added to control the activation threshold.
4. **Summation Function:** Weighted sum of inputs.

$$z = (w_1x_1 + w_2x_2 + \dots + w_nx_n) + b$$
5. **Activation Function:** Decides the output of the neuron (fire or not).

Activation Functions (decide non-linearity)

- **Step function:** If $z > \text{threshold} \rightarrow 1$ else 0
- **Sigmoid:** Smooth curve between 0 and 1
- **Tanh:** Values between -1 and 1
- **ReLU (Rectified Linear Unit):** $\max(0, z)$ (most popular in DL)

Working

1. Inputs are multiplied by weights.
2. Bias is added.
3. Activation function applied.
4. Output sent to next layer.

This mimics “dendrites + soma + axon” of biological neurons.

Biological vs Artificial Neuron

Aspect	Biological Neuron	Artificial Neuron
Unit	Brain cell	Mathematical function
Inputs	Signals from dendrites	Features (x1, x2, ...)
Weights	Synaptic strength	Numerical weights (w1, w2, ...)
Processing	Soma integrates signals	Weighted sum + bias
Threshold	Fires if strong enough	Activation function
Output	Signal via axon	Predicted value (y)
Learning	Synaptic plasticity	Weight adjustment via algorithms (e.g., backpropagation)

Consider below difference between components of Artificial neuron and Human neuron

Component	Human Neuron	Artificial Neuron
Dendrites	Branch-like extensions that receive signals from other neurons.	Input layer — Receives numerical data from features or previous neurons.
Cell Body (Soma)	Processes incoming signals and determines whether to send a signal forward.	Summation function — Adds up weighted inputs.
Axon	Long projection that transmits electrical impulses to other neurons.	Output value — Sent to next layer's neurons.
Synapse	Connection point where signals are passed chemically to another neuron.	Weights — Determine the strength/importance of the connection.